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Nitrogen enrichment and foliar fungal pathogens affect the mechanisms of multispecies plant coexistence

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Summary

- Changes in resources (e.g. nitrogen) and enemies (e.g. foliar pathogens) are key drivers of plant diversity and composition. However, their effects have not been connected to the niche and fitness differences that determine multispecies coexistence.
- Here, we combined a structuralist theoretical approach with a detailed grassland experiment factorially applying nitrogen addition and foliar fungal pathogen suppression to evaluate the joint effect of nitrogen and pathogens on niche and fitness differences, across a gradient from two to six interacting species.
- Nitrogen addition and pathogen suppression modified species interaction strengths and intrinsic growth rates, leading to reduced multispecies fitness differences. However, contrary to expected, we also observed that they promote stabilising niche differences. Although these modifications did not substantially alter species richness, they predicted major changes in community composition. Indirect interactions between species explained these community changes in smaller assemblages (three and four species) but lost importance in favour of direct pairwise interactions when more species were involved (five and six).
- Altogether, our work shows that explicitly considering the number of interacting species is critical for better understanding the direct and indirect processes by which nitrogen enrichment and pathogen communities shape coexistence in grasslands.

Introduction

Nitrogen (N) enrichment is a key global change driver (Stevens *et al.*, 2004; Galloway *et al.*, 2008), which often causes a reduction in plant diversity and changes in species composition (Stevens *et al.*, 2004; Vellend *et al.*, 2017; Midolo *et al.*, 2019). There are several mechanisms by which N can alter plant communities. Adding N may remove a limiting resource for plant growth and thereby decrease the number of niche dimensions (Harpole *et al.*, 2016). This would reduce stabilising niche differences between species and therefore coexistence and diversity (Cenci *et al.*, 2018; Godoy *et al.*, 2018; Losapio *et al.*, 2019). The addition of N may also lead to an increase in fast-growing and tall species (often grasses), which are very competitive for light (Harpole *et al.*, 2016; Seabloom *et al.*, 2017). As light competition is highly asymmetric (Lamb, 2008; DeMalach *et al.*, 2017), a shift towards light limitation is likely to increase competitive differences between species. Hence, N addition has the potential to reduce diversity and change species composition by altering the main classes of processes that maintain coexistence between species pairs. These are the stabilising niche differences that occur when intraspecific exceeds interspecific competition and the fitness differences, which drive competitive exclusion by favouring one competitor over the others, regardless of their abundance. Species are predicted to coexist only when niche differences overcome fitness differences (Chesson, 2000; Saavedra *et al.*, 2017). Changes in niche and fitness differences could also alter species composition if the competitive balance between species changes without coexistence becoming more likely. However, very few studies have empirically assessed how changes in environmental conditions modulate the species interactions that determine coexistence (Bimler *et al.*, 2018; Matías *et al.*, 2018; Hallett *et al.*, 2019). Although many studies have shown that N reduces diversity, we still know little about how it alters these classes of coexistence mechanisms, leading to a large gap in our theoretical knowledge of N enrichment effects. A better understanding of how N affects coexistence mechanisms may allow us to better predict its impact on the diversity and composition of plant communities and unify global change and coexistence research.

In addition to its direct effects, N enrichment can also indirectly impact the diversity and composition of plant communities by altering the abundance of consumers, such as foliar pathogens. N addition will increase the concentration of N in plant tissues, which could benefit pathogens directly (Dordas, 2008), and a shift towards fast-growing species following N addition might also increase pathogen abundance (Cappelli *et al.*, 2020). Changes in pathogen abundance are likely to have further

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consequences for plant coexistence. Pathogens can promote species coexistence through negative density dependence, suppressing species that become common and preventing them from taking over (Petermann *et al.*, 2008; Mordecai, 2011; Bagchi *et al.*, 2014; Bever *et al.*, 2015), thereby increasing stabilising niche differences. Alternatively, leaf fungal pathogens can act as an equalising rather than as a stabilising mechanism by attacking the most dominant species (Mordecai, 2011). For instance, they can reduce the abundance of dominant grasses and thereby reduce competitive asymmetries among dominant and subordinate species in grasslands (Allan *et al.*, 2010; Seabloom *et al.*, 2017). However, other studies have found smaller effects of pathogens: for example, Spear & Mordecai (2018) found a weak relationship between pathogen damage and seed production, suggesting limited effects of fungal pathogens on plant populations. Uricchio *et al.* (2019) also found that fungal infection had weak fitness effects and was unlikely to affect coexistence. Therefore, the efficacy of pathogens as drivers of coexistence remains unclear. Pathogens could also modify the effects of N on coexistence and might offset some of the negative effects of N by reducing species dominance, as shown for insects (Allan & Crawley, 2011). Finally, if pathogen removal and N addition favour different species, then this might lead to complex interactions between the two.

Traditional approaches to understanding coexistence have focussed on pairs of species (modern coexistence theory (Chesson, 2000)). However, it has recently become possible to explore niche and fitness differences between multiple species using a structuralist approach (Saavedra *et al.*, 2017; Song *et al.*, 2018). These advances open now the possibility to investigate how coexistence mechanisms change as we increase the number of species in an assemblage. Theory posits that three or more species can coexist when species interactions create opportunities for coexistence (i.e. structural niche differences) that can accommodate the differences in population growth rates between the species (i.e. the ability of a species to increase its population in the absence of others), so that species vary in abundance over time but none goes extinct (Rohr *et al.*, 2014; Saavedra *et al.*, 2017; Godoy *et al.*, 2018; Fig. 1). Here, the structural fitness differences are a combination of differences in intrinsic growth rates among species and also differences in the position of the feasibility domain, which are determined by the degree of asymmetry in the strength of species interactions (Tabi *et al.*, 2020). Similar to the pairwise scenario, structural niche differences in a multispecies community are partly promoted when a species limits itself more than it limits others (intraspecific competition exceeds interspecific competition; Barabás *et al.*, 2016) but also when indirect interactions provide further opportunities for coexistence. The best studied of these indirect interactions is intransitivity, that is, rock-paper-scissors dynamics in which species A beats species B, species B beats species C, and species C beats species A (Soliveres *et al.*, 2015; Gallien *et al.*, 2017; Godoy *et al.*, 2017). These indirect interactions may also respond to environmental variation, and there is some evidence that intransitivity might be reduced by nitrogen addition (Soliveres *et al.*, 2018). However, there is very little information on how resources and enemies might alter indirect interactions in multispecies communities.

Adding more species to a community might increase the opportunities for indirect interactions and therefore could make coexistence more likely. Conversely, adding more species can hinder rather than promote multispecies coexistence because it increases the chances of including a species with a broad niche that overlaps with the niches of other species (Berlow, 1999; Neutel *et al.*, 2002); and/or augment the chances of including a superior competitor that increases fitness differences (Saavedra *et al.*, 2017). In that case, the multiple species assemblages would need stronger species self-limitation compared with the pairwise case to offset the negative effects of strong and indirect interspecific competition (Barabás *et al.*, 2016; Godoy *et al.*, 2017).

In sum, this means that nitrogen enrichment and pathogens can change the number and the identity of species that can coexist by modifying: species intrinsic growth rates; the strength and sign (both competition and facilitation) of direct interactions among species; and the strength of indirect interactions (e.g. intransitive competition). Experiments manipulating N addition and pathogen infection are required to determine the relative importance of these three mechanisms in affecting multispecies coexistence and how they are modified by nitrogen and pathogens.

In this study, we present results from a grassland biodiversity experiment where nitrogen addition and pathogen suppression, with fungicide, were factorially applied. We quantified the two key pieces of information (species intrinsic growth rates and intra- and interspecific interaction coefficients) required to derive metrics of structural niche and fitness differences among eight perennial plant species. To do so, we measured the growth of individual plants, in terms of biomass change, as a function of the density and composition of their neighbours, in the different experimental treatments (nitrogen, fungicide and their combination; Supporting Information Fig. S1). With this information, we then parameterised a population model to predict which communities could coexist and to characterise the way in which they did so (their structural coexistence metrics). We did this for different numbers of species from two to six. This combination of ecological theory, modelling and detailed field observations allowed us to ask: (1) how do structural niche and fitness differences vary with N addition, pathogen suppression and the interaction between them? (2) How do these effects vary if coexistence between different numbers of species is considered? (3) What is the relative importance of pairwise direct effects and indirect effects arising in multispecies assemblages for coexistence? And (4) what are the predicted consequences of N enrichment and pathogen suppression for biodiversity and species composition?

Materials and Methods

Study system

We conducted this study in the PaNDiv biodiversity experiment, which was set up in October 2015 in Münchenbuchsee (Bern, Switzerland; 47°03′N, 7°46′E, 564 m asl). The mean annual

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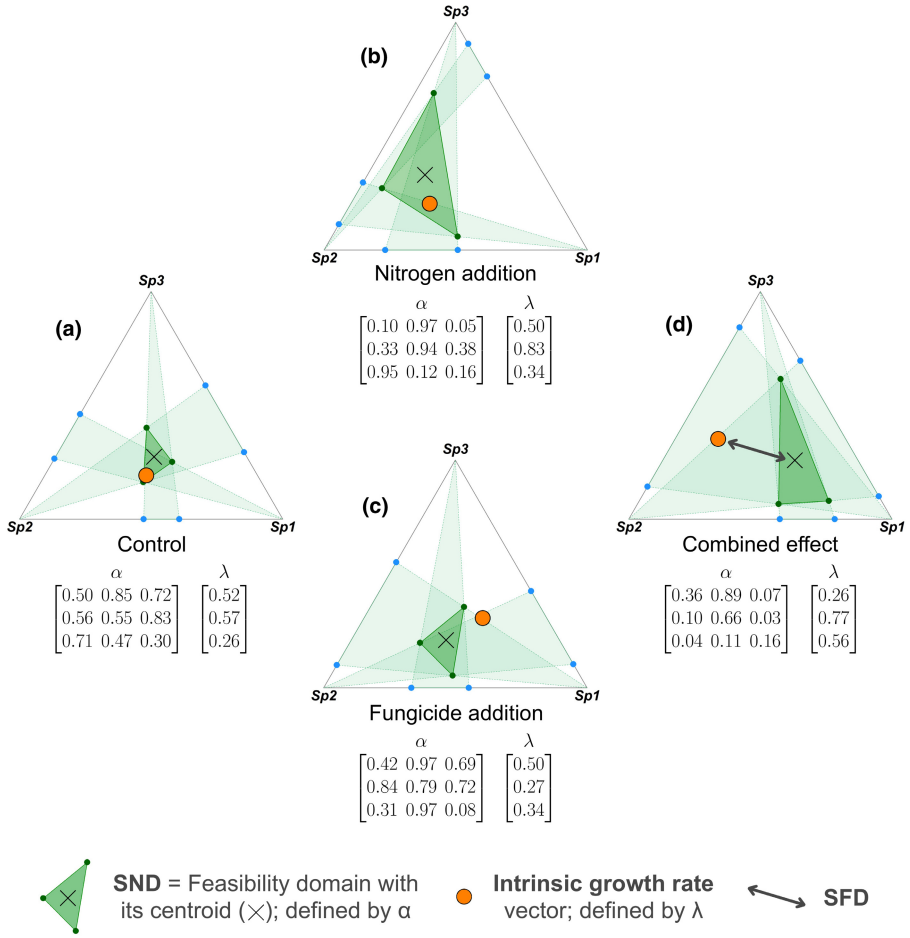


Fig. 1 Structuralist approach (Saavedra *et al*., 2017), applied to understand how environmental changes alter the mechanisms of coexistence between multiple species. Each green triangle is defined by the structure of species interactions (α) and represents the range of intrinsic growth rates that are compatible with species coexistence (i.e. where all species have abundances > 0). Such parameter space is also called the feasibility domain and is shown here for direct pairwise species interactions (light green triangles, defined by the light blue dots) and for the matrix of direct and indirect interactions among three species (dark green triangles, defined by the dark blue dots). This feasibility domain corresponds to the structural analogue of niche differences (SND). Moreover, the distance, displayed as a grey double arrow in panel D, between the centroid of the feasibility domain (×) and the vector formed by the intrinsic growth rates (λ; orange dot) of the species, corresponds to the structural analogue of fitness differences (SFD) for the triplet. Given all this, a triplet can coexist when its feasibility domain can accommodate its intrinsic growth rate vector. Here, we hypothesise that nitrogen and fungicide addition could affect both structural niche and fitness differences. In this example, the control (a) triplet could be feasible despite having small SND if the SFD are also small (the intrinsic growth rate vector is close to the centroid of the feasibility domain). Meanwhile, nitrogen addition (b) is also feasible because of its larger SND, fungicide (c) is unfeasible because SND < SFD, and the combined (d) effect shows not only large SND but also increased SFD, resulting in an unfeasible combination. We show the values of the matrix of species interactions (α matrix) and the vector of intrinsic growth rates (λ vector) that underlies each representation. For a graphical illustration of this approach, we strongly recommend the interactive shiny app by Petry & Lepori (2022) (https://ecodynamics.shinyapps.io/StructuralCoexistence/).

precipitation at the site is 1012.62 mm, and the mean annual temperature is 9.2°C (Federal Office of Meteorology and Climatology MeteoSwiss). The site was formerly an extensively managed grassland that had remained unfertilised since at least 2005, and the field was mown and ploughed to remove the vegetation before the start of the experiment. The PaNDiv experiment consists of 336 experimental plots of 2 × 2 m, where species richness (1, 4, 8 or 20 species) and plant functional composition (fast vs slow-growing species) are manipulated and crossed with applications of nitrogen and fungicide. Communities differing in species number and composition were assembled from a set of 20 common Central Europe perennial grassland species (Table S1). The experiment was arranged in four blocks, with each particular species combination present in each block. The composition of each plot was maintained by manually removing weeds three times a year. The field was mown twice a year, in the middle of June and in August, which corresponds to an intermediate management intensity for lowland grasslands.

The nitrogen and fungicide treatments were applied every year: nitrogen addition plots were fertilised with urea in April and after the first mowing and received in total 100 kg N ha^{-1} yr^{-1} (50 kg ha^{-1} each time). The fungicides Score Profi (Syngenta Agro AG (Basel, Switzerland), 24.8% difenoconazole) and Ortiva (Syngenta Agro GmbH (Maintal, Germany), 22.8% azoxystrobin) were used to suppress foliar pathogens. They were applied four times during the growing season (0.2 ml of Score Profi and

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0.4 ml of Ortiva mixed with 0.062 l of water per plot and time), whereas no-fungicide plots were sprayed with the same amount of water. Each particular combination of species received the four nitrogen × fungicide treatments, to allow us to separate treatment and composition effects. Data from Cappelli *et al.* (2020) showed that the fungicides reduce pathogen incidence by 25.33%, on average, and are particularly effective in reducing rusts and powdery mildews (which were virtually absent from sprayed plots). For further details on the experimental set-up, see Cappelli *et al.* (2020) and Pichon *et al.* (2020).

Estimation of species interactions and intrinsic growth rates across experimental treatments

To study coexistence mechanisms and how they vary across treatments and levels of diversity, we focussed on eight species that had established well at the start of the PaNDiv experiment, thus avoiding environmental filtering processes (which would result in negative intrinsic growth rates). These species are *Taraxacum officinale* Weber (fast-growing), *Crepis biennis* L. (fast) and *Centaurea jacea* L. (slow-growing; Asteraceae), *Rumex acetosa* L. (fast; Polygonaceae), *Dactylis glomerata* L. (fast) and *Anthoxanthum odoratum* L. (slow; Poaceae), *Salvia pratensis* L. (slow; Lamiaceae) and *Plantago media* L. (slow; Plantaginaceae). These are common plant species in temperate grasslands, and the selection involves a mix of grasses and herbs and fast- and slow-growing species. For each species, we measured the biomass of multiple individuals in different competition neighbourhoods to calculate the strength and sign of species interactions and how they varied across nitrogen and fungicide treatments. The natural variation in density and the experimentally imposed relative frequency of these species across plots allowed us to separate and estimate with confidence intra- and interspecific competition as well as species' intrinsic growth rates (see below). We then parameterised a model that describes the population dynamics of competing species, and their likelihood of coexisting, as a function of their network of interactions and their intrinsic growth rates in the absence of competition (see below, structuralist approach). With eight species, we need to estimate 64 pairwise interactions per treatment (eight intraspecific interaction coefficients and 56 interspecific) and a total of 256 pairwise interactions across the four treatments (i.e. control, nitrogen only, fungicide only or both combined).

To account for this high dimensionality, we proceeded as follows. Before the growing season of 2017 (February and early March), we marked focal individuals of each species in neighbourhoods varying in density and relative frequency of co-occurring species, using a bamboo stick together with a coloured plastic tag. We defined the neighbourhood for each focal individual as a 40-cm-diameter circle, marked with a PVC ring during sampling. We selected neighbourhoods in the monoculture plots to estimate intraspecific interactions, and we did the same in the polyculture plots to estimate interspecific interactions (Fig. S1). We measured eight randomly selected neighbourhoods per pairwise interaction, across the four blocks in order to produce gradients of neighbour density (eight neighbourhoods/pairwise interaction × 256 pairwise interactions in the whole experimental design = 2048 neighbourhoods). A minimum cover of at least 5% of either the same or another species was the criterion taken to measure a selected neighbourhood. Those neighbourhoods that did not meet this criterion (because they were dominated by some of the other 12 species in the experiment) were replaced by another one within the same plot. For example, a focal-neighbour combination could be *T. officinale* surrounded by at least 5% cover of *A. odoratum*. It is worth noting that, contrary to previous studies in which polycultures were established containing only pairs of species (e.g. Godoy *et al.*, 2020), we mostly measured polycultures with several different neighbour species (e.g. *T. officinale* surrounded by *C. biennis*, *C. jacea* and *R. acetosa*), as the experiment did not contain two species plots. However, we were able to estimate all pairwise interspecific interactions thanks to the modelling approach applied (see below). This approach is part of the so-called individual fitness models, which are commonly used to estimate interactions between pairs from multispecies data (Bimler *et al.*, 2018; García-Callejas *et al.*, 2020; Martyn *et al.*, 2021). The effect of higher-order interactions (Mayfield & Stouffer, 2017) on coexistence was beyond the scope of this article, and we therefore did not estimate them in our models but our interaction coefficients average across any higher-order interactions that might occur. We visually estimated the cover of the focal species inside the ring as well as the cover of the other species. The cover of the neighbouring species was estimated individually for the eight target species we selected, and we estimated the abundance of all other species grouped into three functional groups (herbs, grasses and legumes). We did not explore the likelihood of coexistence of these groups, but we determined their competitive effects on our eight target species (see statistical models described later).

We measured plant cover at the beginning of the growing season and in June, at peak biomass, before the first cut. This therefore represents the majority of aboveground growth for each plant in that year. We measured plant cover for logistical reasons, but we converted the cover values to biomass for the modelling. To do so, we harvested (and oven-dried, at 60°C for 2 days) the aboveground biomass of eight focal plants per species and treatment, to estimate a relationship between cover and the biomass of each focal species. We obtained reasonably good fits for all and each of the individual species (R^2 = 0.6 ± 0.2, mean ± SD; Fig. S2). Finally, we repeated the whole procedure in 2018 on newly selected focal neighbourhoods, which meant we measured, across the 2 yr, the cover and species composition of 4096 neighbourhoods (2048 each year). Once we obtained data from the experiment, we visually checked for outliers. We took the decision to remove the 2018 data from one species, *Rumex acetosa*, from further analyses because it declined strongly in biomass from February to June during this year (Fig. S3), most likely because conditions were very dry during 2018 and *R. acetosa* had already begun to senesce by June. Modelling this strong decline would indicate that this particular species is not able to grow at the experimental site (i.e. environmental filtering), which is not the case in general.

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We then modelled the interactions between all pairs of species. We used a modified Beverton–Holt function to estimate model parameters (Beverton & Holt, 1957). This modification was applied to model population size as biomass and not as numbers of individuals because the clonal growth of several of these perennial plant species makes it impossible to distinguish separate individuals in the field. However, the model retains the mathematical behaviour of prior work that has shown this function can be applied to both annual and perennial plant species (Cardinaux *et al.*, 2018; Lanuza *et al.*, 2018) and is also suited to apply a structuralist approach (see details in appendix 4 in Saavedra *et al.*, 2017). Since we marked individuals already growing in the field (we did not plant out individuals), we needed to account for the initial biomass of each focal. Focal individuals with greater initial biomass might experience less competition and grow more, as bigger individuals are less sensitive to competition (Rees, 2013). Therefore, we modelled the biomass response (N) of the focal species i in the year $t + 1$ as follows:

$$\frac{N_{i,t+1}}{N_{i,t}} = \frac{\lambda_i \left(1 + \theta_i N_{i,t}\right)}{1 + \sum_{j=1}^{n} \left(\alpha_{ij} + \Omega_i N_{i,t}\right) N_{j,t}} \qquad \text{Eqn 1}$$

where λ_i is the per capita growth rate measured as biomass of i in the absence of interactions and α_{ij} represents the effect that each neighbour species j has on i. The parameters θ_i and Ω_i reflect the effect of the initial size of species i on its intrinsic growth rate (λ_i) and its interactions with neighbour species (α_{ij}), respectively. We fitted Eqn 1 to empirical observations using maximum likelihood methods (optimx function within the OPTIMX package (v.2022-4.30, Nash & Varadhan, 2011); optimx method = 'Nelder-Mead'). We estimated λ_i (bounded to be positive as negative intrinsic growth rates for species growing alone do not make sense given that all species can persist at the site) and α_{ij} for each treatment separately (control, N addition, fungicide and combined). α_{ij} is not bounded and can be negative (competition) or positive (facilitation) because the structuralist approach can compute niche differences when both types of interactions are combined. We fitted three different models for each species to deal with initial biomass effects. We considered that initial biomass (Ω) had: no effect on the other model parameters (i.e. the effect of size is negligible); a common size effect across all species; or a species-specific size effect. To identify the best model for each species and treatment, we used AIC (Burnham & Anderson, 2002). All the details about the three different specifications of the function, and their AIC estimates for each species, are included in Table S2. Note that this modelling procedure was also done separately for the years 2017 and 2018. To better understand the confidence of our results, we calculated the Hessian matrix with the ghgen function within the OPTIMX package and transformed its diagonal to obtain the standard error for all model parameter coefficients (95% confidence), assuming that the gradient is well approximated by a quadratic function. Mean model estimates and standard errors for the α_{ij} and λ_i parameters under each treatment are provided in Tables S3–S6.

Predicting effects on N enrichment on species coexistence with the structuralist approach

With the field parametrisation of the dynamic model describing the population dynamics of interacting species, we applied the structuralist approach to explore how coexistence mechanisms varied across the experimental treatments. We followed previous definitions of Saavedra *et al.* (2017) to compute metrics of structural niche differences (SND), structural fitness differences (SFD) and feasibility (i.e. an equilibrium point which is assumed to be globally stable following previous approaches (Zhao & Luo, 2010; Saavedra *et al.*, 2017); thus, here we consider a feasible combination to be one in which all species coexist). We did this for species pairs and for combinations of three to six species. The great advantage of the structuralist approach is that we can evaluate the probability of coexistence for any number of species at the same time if we have information on their intrinsic growth rates and the full matrix of pairwise interactions. For all species combinations, coexistence conditions are the same: a species combination can coexist when SND are greater than SFD (Fig. 1). Importantly, we can compare the variation in SND and SFD across treatments, but we cannot compare them across combinations of different numbers of species, that is, SND in a three-species combination (triplet) cannot be compared with the SND in a four-species combination. This is because SND and SFD inherently vary as more species are included in the combination, as indirect interactions between species contribute to SND and these vary with the number of species included. We also evaluated the importance of indirect interactions (e.g. intransitivity and other types of species interaction chains) in allowing species to coexist using the metric called community-pair differential (Saavedra *et al.*, 2017). Community-pair differential varies between −1 and 1. A positive value indicates that indirect interactions promote coexistence, that is, there are greater opportunities for coexistence in the multispecies combination than in the individual pairs that make up the combination. For example, in the case of an intransitive triplet, no individual pair could coexist, but the triplet could coexist. Negative values of the differential indicate the opposite, that is, fewer opportunities for coexistence in the multispecies combination. These computations were performed for the mean model parameters, the lower and upper tails of the 95% confidence intervals, and for the years 2017 and 2018 separately.

We finally used generalised linear models (GLMs) to evaluate the effect of N addition and fungicide application on these coexistence metrics, for each number of species in the combination. To determine what distribution to use for the residuals in the analysis of SND, SFD, community-pair differential and feasibility metrics, we used the function descdist within the package 'fitdistrplus' (Delignette-Muller & Dutang, 2015), which led us to fit a GLM with gamma errors (link = 'identity') for SND, SFD and community-pair differential metrics and binomial errors for the feasibility metric. A summary of the results from the different models for all metrics is available in Tables S7–S26. All computational analyses were performed in R v.4.2.1 (R Core Team, 2022).

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Results

Determinants of feasibility across treatments

Our experimental treatments resulted in communities with a wide range of structural niche differences (SND) and structural fitness differences (SFD). Nevertheless, the variation was higher for SFD than for SND regardless of the number of species considered. Accordingly, most communities were predicted to be unfeasible and not able to coexist (Fig. 2). We did not observe a clear threshold between feasible and unfeasible assemblages in multispecies communities mainly because indirect interactions promoted species persistence for some species combinations and hindered it for others. For instance, feasible communities were those in which we found: intraspecific strongly exceeded interspecific interactions; quasineutral-like dynamics in which all interactions and growth rates were equivalent but intraspecific interactions were slightly stronger than interspecific ones; and intransitive dynamics. Conversely, examples of predicted unfeasible communities were those in which interspecific interactions were stronger than intraspecific interactions (i.e. priority effects); and high variation in intrinsic growth rates among species destabilised the effect of niche differences.

Effects of nitrogen enrichment and fungal pathogen suppression on structural niche and fitness differences

The effect of nitrogen (N) addition, pathogen suppression and their interaction on changing SND compared with the control varied depending on the number of interacting species (Fig. 3a). Within species pairs, N addition and pathogen suppression increased SND when combined but did not differ from control conditions when applied alone. N addition slightly decreased SND in three-species combinations; however, the combination of N addition and pathogen suppression increased SND. For assemblages of four, five and six species, the pattern was more consistent: N addition, pathogen suppression and their combination tend to increase SND compared with control conditions. Separating each year individually, we observed in 2017 that N addition, pathogen suppression and their combination decreased SND. This result differs from both years combined, and such differences are more pronounced with a higher number of interacting species (Fig. S4a). Analysing 2018 alone showed similar results to both years combined (Fig. S4b). We also explored how our experimental treatments modified SND at the lower and upper limits (95% CI), which rendered smaller and less variable SND values than the mean SND for both years (Fig. S5).

On the contrary, SFD did not vary between treatments and control conditions for combinations of two species whereas for assemblages of three species only the combination of N addition and pathogen suppression slightly increased SFD. Similar to the case of SND, we observed parallel trends from four to six-species assemblages. Specifically, N addition, pathogen suppression and their combination strongly decreased SFD (Fig. 3b). The separated analyses of SFD in 2017 and 2018 showed contrasting results. In 2017, the reduction in SFD compared with control conditions is repeated for assemblages of four, five and six species (Fig. S6a). However, we observed the opposite pattern in 2018; namely, all treatments increased SFD in assemblages of four-, five- and six-species combinations (Fig. S6b). Despite the contrasting results, analysing both years individually confirms the general pattern that variation of SFD differences between treatments was highly dependent on the number of interacting species. Finally, the exploration of how our experimental treatments modified the lower and upper limits (95% CI) of SFD showed smaller and more variable values but comparatively consistent with the mean values for both years, particularly in the upper limit (Fig. S7).

The role of indirect effects

Both direct pairwise interactions and multispecies indirect interactions can affect the structural niche differences (SND), and the relative contribution of each interaction type can be evaluated by the community-pair differential. Overall, we observed small values of community-pair differential (i.e. on average across treatments, only 15% of the SND were due to indirect interactions, for three-species combinations), and these community-pair differential values tended to zero with increasing species richness. Contrary to our initial expectation, this result suggests that indirect interactions have a minor role in promoting SND as more species are added (Fig. 4). Equally surprising is that we did not observe differences when specific treatments were compared. When decomposing the main results into 2017 and 2018, we observed the same overall decay in the importance of indirect effects with the number of interacting species and experimental treatments had no effect (Fig. S8). The exploration of the lower and upper limits (95% CI) of community-pair differential values corroborated the low importance of indirect interactions on promoting species coexistence because all values tended to zero in the lower limit regardless of the number of species and treatment considered (Fig. S9a), and upper limits reproduced the decaying importance of indirect interactions for promoting SND as the number of species increased (Fig. S9b).

Consequences for diversity maintenance and changes in species composition

The combination of SND and SFD resulted in a reduced proportion of feasible species combinations as the number of interacting species increased (Fig. 5). This result is particularly apparent for the control treatment, which does not sustain any feasible combination of six species. However, the nitrogen and fungicide treatments increased the number of feasible combinations when four or more species were considered. When we separated both years, we observed a more abrupt decay in the proportion of feasible combinations with the number of interacting species, particularly in 2018 when there are no feasible combinations of five or six species (Fig. S10). In 2017, the fungicide treatment seems to support the most feasible combinations, especially in combinations of three and four species. The exploration of the lower and upper limits (95% CI) in both years combined offered a substantial

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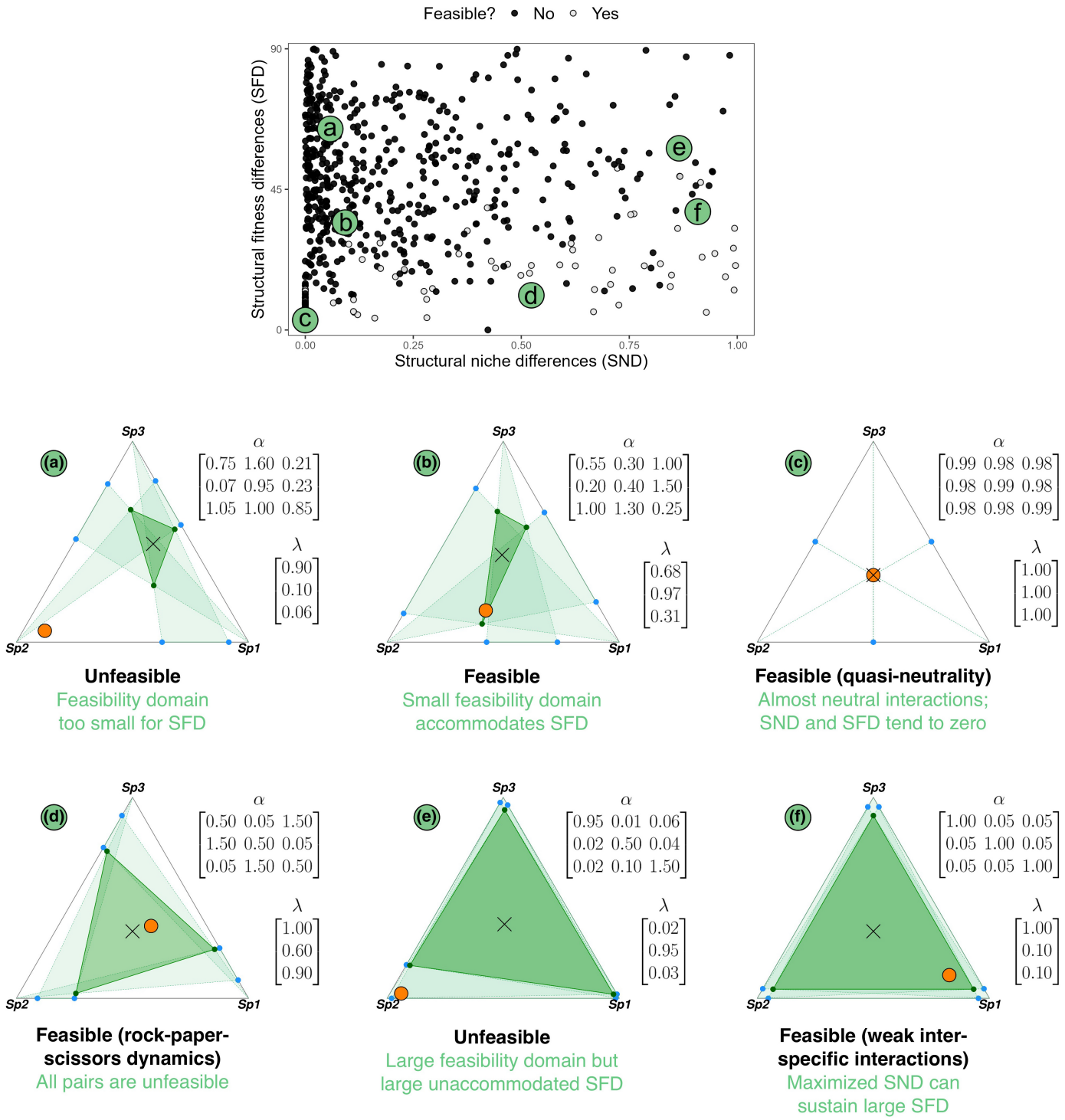


Fig. 2 Diagram describing the relationship between the structural niche differences (SND) and the structural fitness differences (SFD) for all three-species combinations across treatments in our study, together with some examples of how these mechanisms relate to the underlying network of interaction coefficients. Data points in black and white represent unfeasible and feasible triplets of species, respectively. The triangles visually represent the structuralist approach (Saavedra *et al*., 2017). Coexistence is achieved when the feasibility domain or SND (green triangles with blue dots, defined by the structure of species interactions (α)) are capable of accommodating the vector of intrinsic growth rates (λ; orange dot). The distance between λ and the centroid of the feasibility domain (×) visually approximates the SFD. Low values of SND usually generate unfeasible combinations (a), although even some small feasibility domains can have shapes that may accommodate intermediate values of SFD (b) due to the structure of the interaction network. Moreover, quasi-neutrality (c) is a particular example of a feasible combination where both SND and SFD tend to zero. Intransitive dynamics such as rock-paper-scissors (d) may also occur, in which the multispecies combination is feasible but all the pairwise interactions are unfeasible. Finally, even assemblages with high niche dimensionality can be unfeasible in combination with specific intrinsic growth rates (e), although the most likely scenario is that these intrinsic growth rates can be accommodated and a feasible assemblage produced (f). We show the values of the matrix of species interactions (α matrix) and the vector of intrinsic growth rates (λ vector) that underlies each representation.

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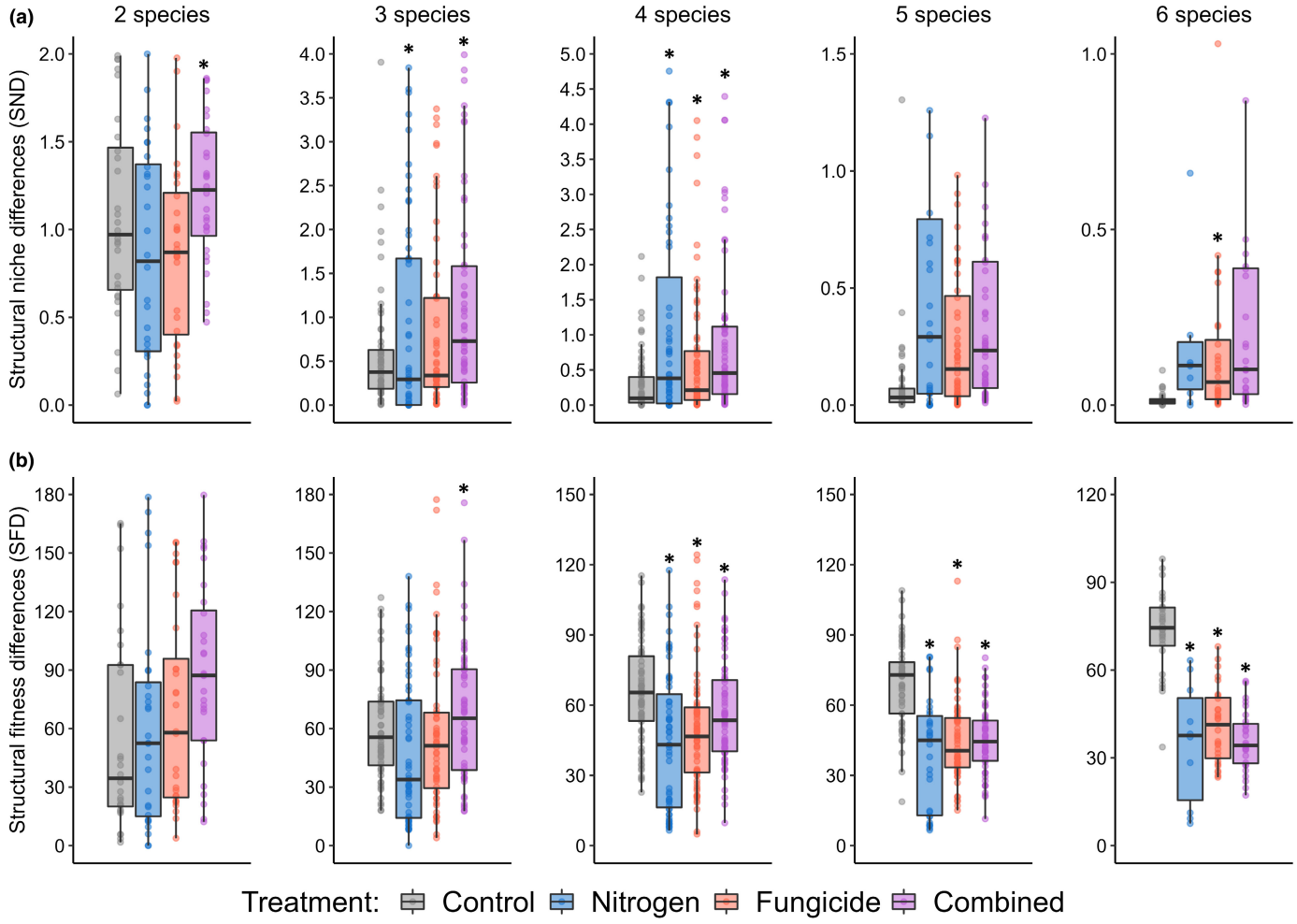


Fig. 3 Structural niche (a) and fitness (b) differences for all combinations of two to six species, in the four experimental treatments (control, nitrogen, fungicide and combined). Each point represents a value of SND or SFD of a specific species combination under a particular treatment. Lines within boxes represent the median. Asterisks above boxes indicate significant differences (GLM, $P < 0.05$) between the corresponding treatment and the control (Supporting Information Tables S7, S8). Note that the axis scales differ between panels because results can be compared across treatments within each species combination but not between species combinations (Saavedra *et al*., 2017). Results represented correspond to mean estimates of interaction coefficients and intrinsic growth rates when data from years 2017 and 2018 were combined. For mean values of each year separately, the same representation and statistical analyses can be found in Fig. S4 and Tables S9, S10 (SND) and Fig. S6 and Tables S11, S12 (SFD). For the lower and upper limits (95% CI) of both years combined, the same representation and statistical analyses can be seen in Fig. S5 and Tables S13, S14 (lower limit) and in Fig. S7 and Tables S15, S16 (upper limit).

decrease in feasibility with almost no predictions of feasible combinations, especially for multispecies assemblages (Fig. S11).

Together with the impact on the overall numbers of species predicted to coexist, our treatments did predict strong compositional changes. For instance, only five species were present in all species combinations under control conditions, with *T. officinale* and *A. odoratum* being unable to coexist in five- and six-species combinations and *P. media* in six-species combinations (Fig. 6a). Conversely, most species were evenly distributed across all feasible species combinations under the nitrogen addition treatment, with the exception of *A. odoratum* which was unable to coexist in any assemblages of six species (Fig. 6b). Pathogen suppression also had large effects on composition with some species such as *C. jacea* struggling to coexist with others in the fungicide treatment and others such as *D. glomerata* expected to coexist more frequently, especially in assemblages of three or more species (Fig. 6c). Finally, the combined treatment of nitrogen addition and pathogen suppression had mixed effects: some species such as *D. glomerata* and *S. pratensis* were unable to coexist in some cases, but it was the only treatment in which all eight species had opportunities for coexistence in six-species assemblages. Altogether, we observed that different species benefited from different treatments and that these outcomes were highly dependent on the number of interacting species in a given combination.

Discussion

Nitrogen enrichment is a major driver of biodiversity declines; however, it is not clear whether it does so by altering niche or fitness differences between species, or by changing the nature of

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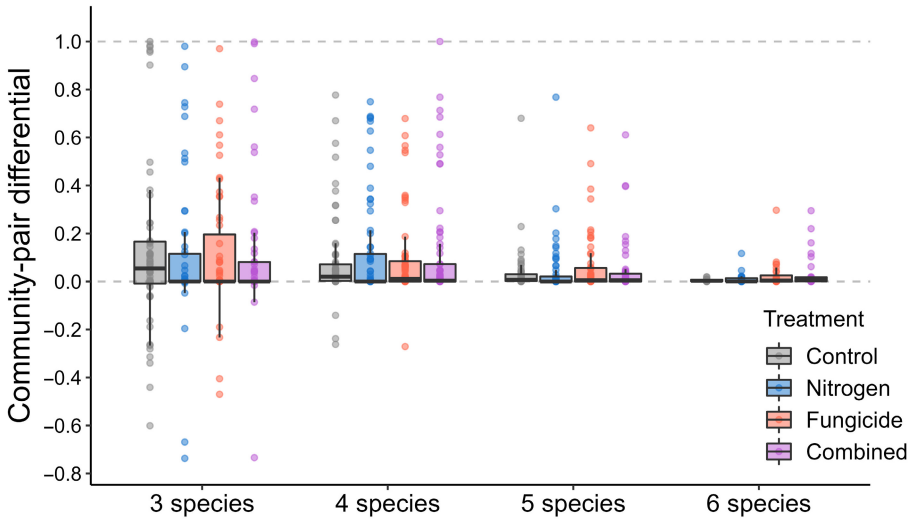


Fig. 4 Community-pair differential for combinations from three to six species simultaneously, in the four experimental treatments (control, nitrogen, fungicide and combined). Each point represents a value of community-pair differential of a specific species combination under a particular treatment. Lines within boxes represent the median. Asterisks above boxes indicate significant differences (GLM, $P < 0.05$) between the corresponding treatment and the control (Supporting Information Table S17). Results represented correspond to mean estimates of interaction coefficients and intrinsic growth rates when data from years 2017 and 2018 were combined. For mean values of each year separately, the same representation and statistical analyses can be found in Fig. S8 and Tables S18, S19. For the lower and upper limits (95% CI) of both years combined, the same representation and statistical analyses can be seen in Fig. S9 and Tables S20, S21.

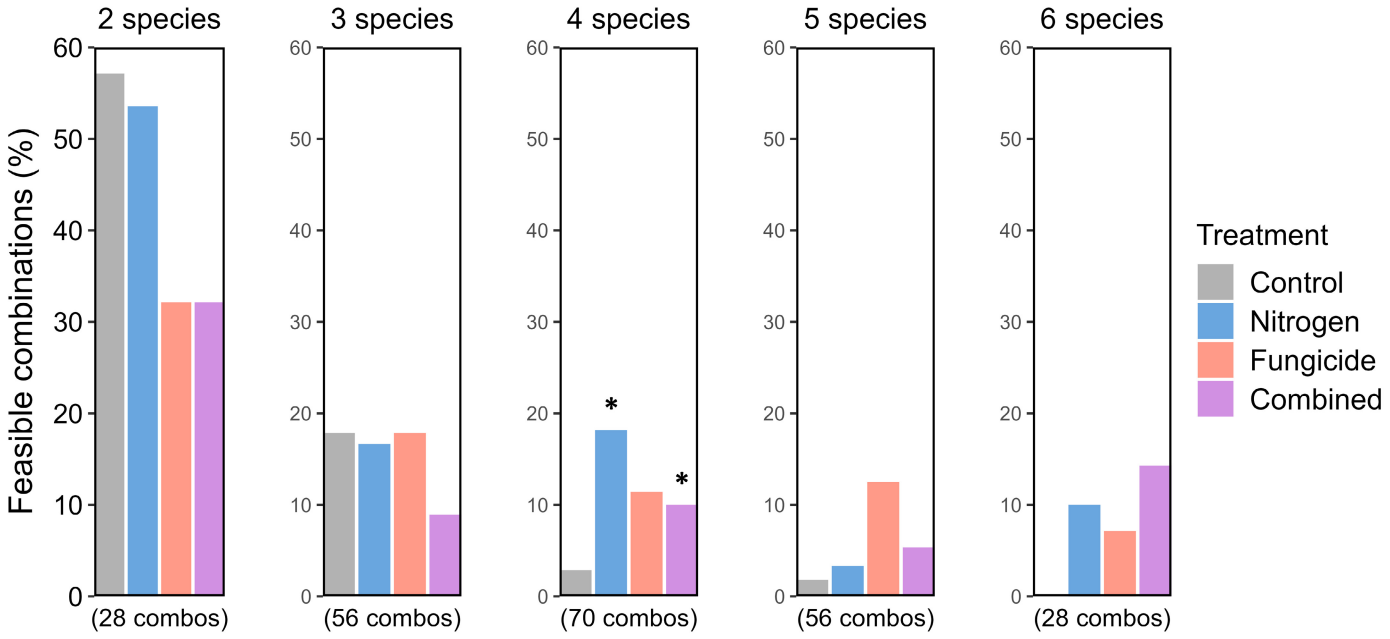


Fig. 5 Feasible combinations (% of the total) of two to six species, in the four experimental treatments (control, nitrogen, fungicide and combined). A feasible combination is one in which all species are predicted to coexist. Asterisks indicate significant differences (GLM, $P < 0.05$) between the treatments and the control (Supporting Information Table S22). The total number of possible combinations (combos) of two–six species is shown at the bottom. Results represented correspond to mean estimates of interaction coefficients and intrinsic growth rates when data from years 2017 and 2018 were combined. For mean values of each year separately, the same representation and statistical analyses can be found in Fig. S10 and Tables S23, S24. For the lower and upper limits (95% CI) of both years combined, the same representation and statistical analyses can be seen in Fig. S11 and Tables S25, S26.

indirect interactions emerging in multispecies assemblages. We used recent advances in ecological theory and parameterised models describing the dynamics of competing species with data from a controlled field experiment to explore how N addition altered coexistence mechanisms in assemblages of different numbers of species. We observed that experimental applications of N tended to increase the strength of stabilising niche differences between plant species. This finding aligns with a recent global-scale study by Band *et al.* (2022) that did not find much evidence that N reduced niche dimensionality. Although the

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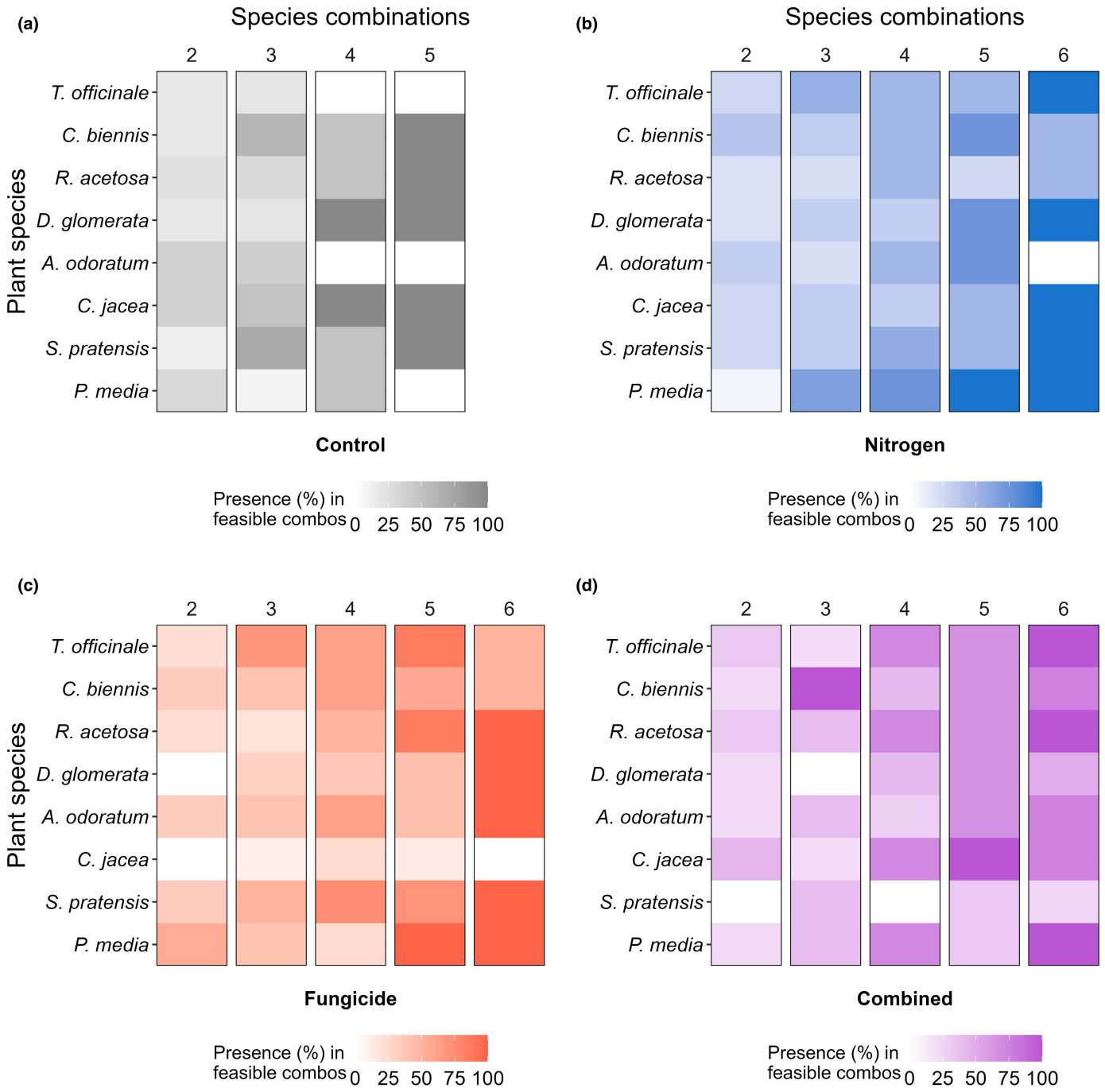


Fig. 6 Probability (%) of each of the eight plant species to stably coexist with their neighbours in combinations from two to six species simultaneously. Results are shown for the four treatments: (a) control, (b) nitrogen, (c) fungicide and (d) combined. The 6-species column has been removed from the control panel because no combinations were predicted to be feasible. Bear in mind that a particular species may not coexist either because it is competitively dominant or subordinate. Results represented correspond to mean estimates of interaction coefficients and intrinsic growth rates when data from years 2017 and 2018 were combined. Species in the vertical axis are *Taraxacum officinale* Weber, *Crepis biennis* L., *Centaurea jacea* L., *Rumex acetosa* L., *Dactylis glomerata* L., *Anthoxanthum odoratum* L., *Salvia pratensis* L. and *Plantago media* L.

addition of N is expected to reduce the number of niche dimensions and opportunities for trade-offs in the uptake of different nutrients to promote coexistence (Tilman, 1982; Harpole *et al.*, 2016), plant responses to a higher N availability depend not only on the quantity of N but also the chemical form (Bobbink *et al.*, 2010). The mineralisation of urea (CH_4N_2O) in the soil produces different N forms at varying rates that may be assimilated differently depending on the plant species (Kahmen *et al.*, 2006; Harrison *et al.*, 2007). Particularly, ammonium (NH_4^+) and nitrate (NO_3) are most frequently used (Hachiya & Sakakibara, 2017), but ammonia (NH_3) (Miflin & Lea, 1976) and even urea itself may be uptaken under certain conditions

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(Witte, 2011). Therefore, increases in different N sources may promote stabilisation (e.g. McKane *et al.*, 2002; Miller & Bowman, 2002), especially if these N forms were not limiting in the initial control conditions of the experiment (Harpole *et al.*, 2016).

Our study also finds that N enrichment decreases fitness differences between species. This result contradicts a large body of literature that has described how higher levels of N result in species loss by increasing plant productivity and selecting for fast-growing, light-competitive species (Suding *et al.*, 2005; Clark & Tilman, 2008; Bobbink *et al.*, 2010; Storkey *et al.*, 2015; DeMalach *et al.*, 2017; Stevens *et al.*, 2018). A possible explanation could be that N addition increased the competitive ability of smaller, nutrient-limited species relative to those which are bigger or not limited by nutrients (Schwinning & Weiner, 1998; Goldberg *et al.*, 2017) and thereby equalised intrinsic growth rates and competitive ability across plant species. Additionally, if plants remained limited by resources such as water or phosphorus, we may not have seen a large increase in asymmetric light competition following N addition because plants still competed strongly for these resources. Overall, this study suggests that N enrichment may be able to promote niche differentiation via modifications in the structure of direct and indirect interactions. This allows imbalances in intrinsic growth rates (i.e. the ability of species to grow in the absence of competition) to be better accommodated and therefore promotes coexistence.

In addition, our experiment showed that natural enemies such as leaf pathogens have contrasting effects on natural communities. We found that the suppression of pathogen communities produced the same outputs as observed for N enrichment. That is, increases in structural niche differences and decreases in fitness differences. One possibility is that fungicide changed the composition of pathogen communities by reducing rusts and mildews but not leaf spots (Cappelli *et al.*, 2020). These changes in leaf pathogen communities could have selected more aggressive and generalist pathogens that equalise fitness differences between plants and promote niche differences via negative density dependence. In addition, reducing fungal pathogen abundance promoted the fast-growing plant species (Cappelli *et al.*, 2020) and this may have equalised their competitive ability relative to the slow species, which are otherwise dominant at our site.

By applying a structural approach (Saavedra *et al.*, 2017) to mechanistically understand species coexistence, we found two novel results. First, the modifications in niche and fitness differences caused by N addition, pathogen suppression and their combination differed between pairs of species and multispecies assemblages (three to six species). This finding shows that the effects of global change drivers on coexistence are contingent upon the number of interacting species, and therefore, suggests that changes in coexistence mechanisms may be complex as diversity is lost progressively from communities. In addition, increasing N or suppressing pathogens may have different effects depending on the initial diversity of the community. Second, comparing the importance of direct pairwise interactions against indirect interactions suggests that indirect interactions provide new opportunities for multispecies coexistence in perennial grasslands. For assemblages of three species, we observed a wide variety of indirect interactions including both intransitive competition (i.e. rock-paper-scissors dynamics) and indirect facilitation (i.e. a superior competitor harms an intermediate competitor more than a weak competitor and indirectly promotes the persistence of the weakest competitor; Soliveres *et al.*, 2015; Godoy *et al.*, 2017). However, indirect interactions lost importance for promoting coexistence as the number of interacting species increased. With five or six species, indirect interactions almost did not increase the opportunities for coexistence compared with direct interactions (i.e. did not increase structural niche differences). This result may arise due to a sampling effect: when more species are considered, the effect of intraspecific competition on promoting coexistence is stronger than the effect of indirect interactions. This result suggests that processes such as extinction cascades are less likely to destabilise multispecies communities but could accelerate diversity loss once it has already started to decline to lower levels. Overall, both novel results indicate a strong synergy among environmental drivers and the number of interacting species. Further research on this synergy can give more insights into how direct and indirect interactions shaped by complex combinations of biotic and abiotic factors can modulate the effect of global change drivers on biodiversity maintenance.

When looking at predictions of biodiversity maintenance across our experimental treatments, the main pattern we observed was a reduction in the number of feasible combinations as the number of interacting species in the community increased. This pattern appeared because larger assemblages were unable to accommodate the differences in intrinsic growth rates between species, that is, they were more likely to include a dominant competitor, which would drive other species to extinction. Control conditions maintained the highest proportion of species pairs (57% feasible pairs) compared with N addition, pathogen suppression and their combination (40% feasible pairs). However, we did not predict any feasible combination under control conditions for six-species assemblages whereas the treatments were predicted to maintain 10% of six-species combinations. Altogether, this means that our global change treatments reduced diversity as expected for assemblages with low numbers of interacting species but reversed the pattern for larger assemblages. Apart from these predicted declines in species richness, we found support for changes in species composition under each treatment. Some particular species illustrate these compositional changes. For instance, *T. officinale* could coexist in most combinations under N addition because its growth rate was reduced in comparison with control conditions, as previously observed (Murray *et al.*, 1983). Also, *C. jacea* could not coexist under pathogen suppression perhaps because it was one of the species that most strongly increased its pathogen infection in mixed communities, presumably due to spillovers from other species (Cappelli *et al.*, 2022). Finally, *D. glomerata* under N addition and *P. media* under pathogen suppression did not coexist in species pairs but could coexist in multispecies assemblages, indicating that the compositional consequences of N addition, pathogen suppression and their combination are also dependent on the number of interacting species.

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It is worth recalling that these predictions assume that environmental conditions, either N enrichment or pathogen suppression, do not vary in space or time. However, environmental variation is pervasive in ecological communities and can promote species coexistence (Adler *et al.*, 2006; Sears & Chesson, 2007). For example, the combination of lower dispersal rates and more patchy landscapes with high environmental heterogeneity help increase coexistence in larger areas by partitioning space (i.e. increase in niche differences) and overcoming competitive imbalances between species (Hart *et al.*, 2017; García-Callejas *et al.*, 2021). Similarly, temporal variation can help species to partition the environment and increase coexistence opportunities (Hallett *et al.*, 2019). In fact, thanks to our two-year sampling effort we were able to capture how temporal variation affects the mechanism of biodiversity maintenance. For instance, N addition, pathogen suppression and their combination tended to decrease fitness differences in 2017 but increase them in 2018 (Fig. S6). Future research therefore needs to span the spatial and temporal scales to study which component, either niche or fitness differences, is more variable. Another important point not covered in the present study is that we have assumed that interactions only occur between adult plants. This means that our study ignores the contribution of other life-history stages such as germination, recruitment or seedling competition, which influence establishment success in perennial grasslands (Peart, 1989; Tilman, 1997; Jakobsson & Eriksson, 2000) and, thus, the ability of species to persist and coexist with others.

With increasing evidence of the challenges that global change poses for biodiversity, it is of the utmost importance to gain a mechanistic understanding of how global change drivers such as nitrogen enrichment shape species interactions and coexistence processes (Tylianakis *et al.*, 2008). In this study, we show that N enrichment reduces species coexistence at the pairwise level but may promote coexistence at the multispecies level by increasing niche and decreasing fitness differences between plants. We also document contrasting effects of fungal pathogens on direct and indirect interactions, with pathogens hindering coexistence by reducing niche and increasing fitness differentiation, suggesting that an increase in pathogen abundance following N enrichment would have complex effects on diversity change. The novel finding that the more interacting species there are in a community, the more likely biodiversity is maintained under a combination of N enrichment and pathogen suppression (thanks to increasing niche differences and decreasing fitness differences) suggests that changes in resources and enemies may have complex effects on coexistence. Altogether, these results highlight the importance of mechanistically understanding how N enrichment, directly and indirectly, alters biodiversity via modifications in the architecture of multispecies interactions.

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Competing interests

None declared.

Author contributions

EA set up the PaNDiv experiment; RRG, EA and OG designed the research and the statistical analyses; RRG performed the research, analysed the data and built the figures; and RRG wrote the manuscript with input from OG and further revisions from all authors.

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Data availability

Data and code used to generate the results of this manuscript are publicly available in Zenodo (doi: 10.5281/zenodo.7343389).

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Sampling method through the PaNDiv experiment.

Fig. S2 Relationships between plant biomass and cover.

Fig. S3 Change in biomass after the growing season.

Fig. S4 Structural niche differences for 2017 and 2018 separately.

Fig. S5 Structural niche differences in the lower and upper confidence limits.

Fig. S6 Structural fitness differences for 2017 and 2018 separately.

Fig. S7 Structural fitness differences in the lower and upper confidence limits.

Fig. S8 Community-pair differential for 2017 and 2018 separately.

Fig. S9 Community-pair differential in the lower and upper confidence limits.

Fig. S10 Proportion of feasible combinations for 2017 and 2018 separately.

Fig. S11 Proportion of feasible combinations in the lower and upper confidence limits.

Table S1 Full list of species used in the PaNDiv experiment.

Table S2 Selected population models for each treatment and focal species.

Table S3 Alpha and lambda coefficients for the control treatment.

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Table S7 GLM results for structural niche differences across years.

Table S8 GLM results for structural fitness differences across years.

Table S9 GLM results for structural niche differences in 2017.

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Table S13 GLM results for structural niche differences in the lower confidence limit.

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Table S16 GLM results for structural fitness differences in the upper confidence limit.

Table S17 GLM results for community-pair differential across years.

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Table S22 GLM results for feasibility across years.

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